Module 2 solution

```
source("R/prepare_everything.R")
```

```
[1] "Metadata imported!"
[1] "Chronos scores imported!"
```

```
[1] "Expression data imported!"
[1] "Mutation data imported!"
```

Chronos scores: Gene essentiality

Genes can be relevant or irrelevant for cellular survival. Essential genes strongly impact cellular growth if they are perturbed. Chronos scores in DepMap quantify how strongly a cancer cell line depends on a particular gene for its growth or survival.

The key principle is: **the more negative the Chronos score, the stronger the dependency on that gene**.

Chronos score	Interpretation
+0.5	Gene knockout may confer a growth advantage
0	No substantial effect on cell growth or survival
-0.5	Moderate dependency
-1	Effect comparable to the median common-essential gene
-2	Very strong dependency

2.1

Explore the `chronos` dataset. What does it contain? How many genes are generally essential (have a mean chronos score of < 1 across all cell lines)?

```
sum(apply(chronos, 2, mean, na.rm = TRUE) < -1)
```

```
[1] 876
```

2.2

Familiarize yourself with the `mut` mutation dataset. How many cell lines show a mutation for each of the oncogenes? What is the most frequently mutated oncogene?

```
mut[HugoSymbol %in% oncogenes, .N, by = HugoSymbol][order(-N)]
```

```
    HugoSymbol     N
        <char> <int>
 1:       TP53   923
 2:       KRAS   223
 3:        APC   192
 4:       PTEN   172
 5:     CDKN2A   171
 6:    SMARCA4   162
 7:     PIK3CA   161
 8:       BRAF   139
 9:        RB1   132
10:       EGFR    82
11:      BRCA1    77
```

2.3

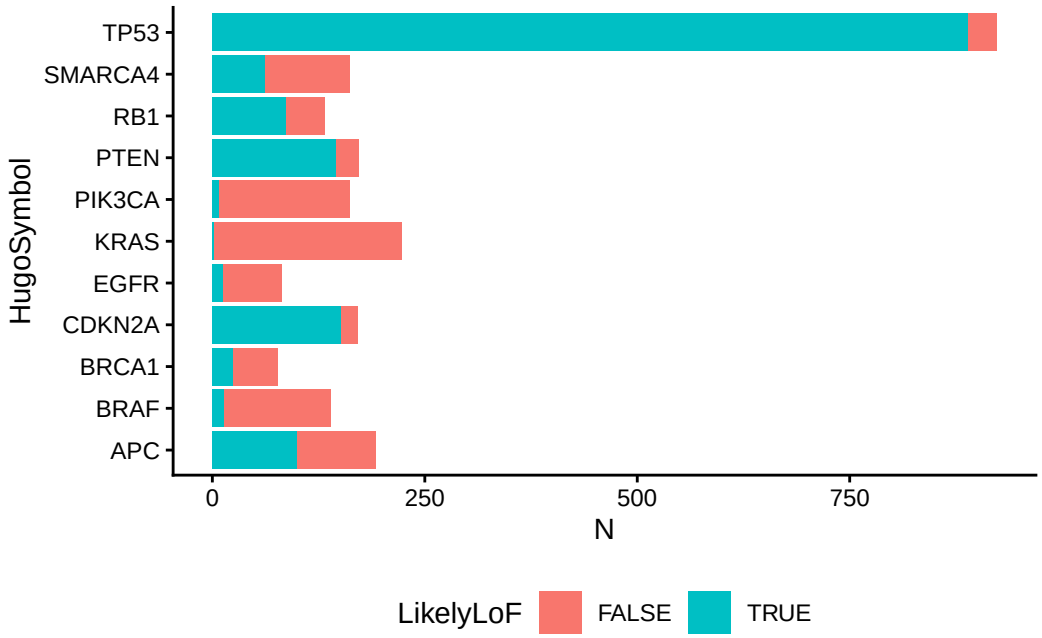
`oncogenes` contains a set of ten genes that are strongly associated with tomorigenesis and therefore highly clinically relevant.

```
oncogenes
```

```
 [1] "TP53"    "KRAS"    "BRAF"    "BRCA1"   "APC"     "PTEN"    "PIK3CA"
 [8] "EGFR"    "RB1"     "CDKN2A"  "SMARCA4"
```

Tumors can be caused by different kinds of mutations. Popular tumor-causing mutations are loss of function of tumor suppressor genes. Compute the number of mutations per oncogene, now also including if the mutation is likely a loss of function mutation. Find a way to plot these numbers. Do you observe a difference for TP53 and KRAS?

```
ggplot(mut[HugoSymbol %in% oncogenes, .N, by = .(HugoSymbol, LikelyLoF)],
  aes(N, HugoSymbol, fill = LikelyLoF)) +
  geom_col() +
  theme(legend.position = "bottom")
```



2.4

TP53 encodes p53, which can trigger cell-cycle arrest or apoptosis. MDM2 negatively regulates p53. In a cell with functional p53, removing MDM2 can reactivate p53 signaling and strongly impair cellular growth. In a cell where TP53 is functionally inactivated, removing MDM2 often has a much smaller effect.

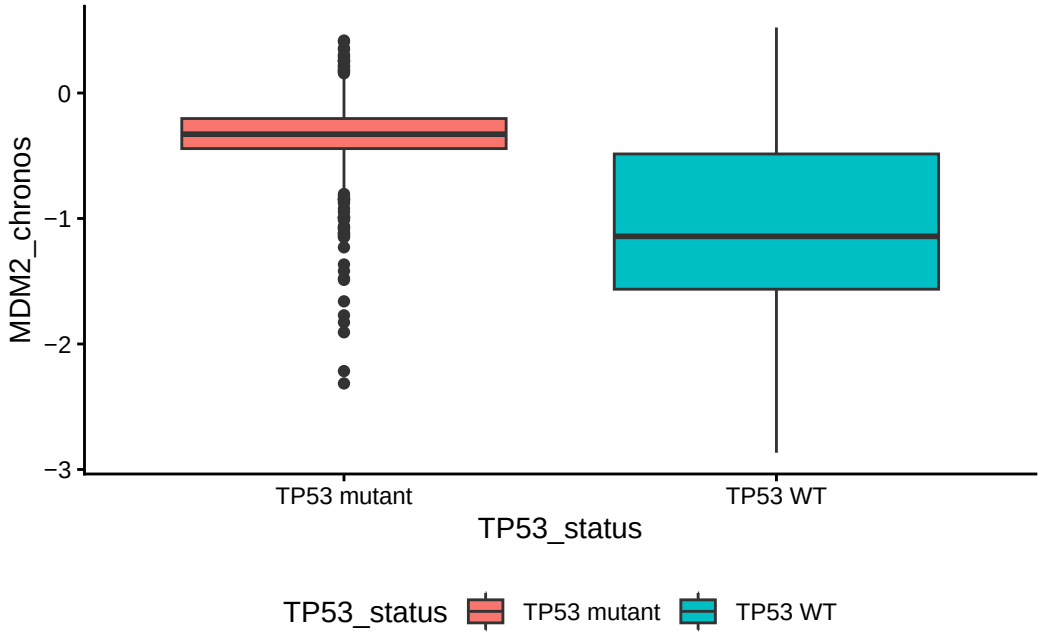
Are TP53-wild-type cell lines more dependent on MDM2 than TP53-mutated cell lines? We want to explore if there is an observable difference in the Chronos scores of MDM2 for cell lines with a mutation in TP53. Find out which cell lines likely have a disfunctional / inactive TP53 gene. Explore if the Chronos scores of MDM2 are different for these cell lines compared to cell lines with functional TP53. Optionally, find a way to visualize that.

```
tp53_mut_cell_lines <- mut[HugoSymbol == "TP53" & LikelyLoF, unique(ModelID)]

plot_data <- data.table(
    cell_line = rownames(chronos),
```

```
    TP53_status = fcase(rownames(chronos) %in% tp53_mut_cell_lines, "TP53 mutant",
    default = "TP53 WT"),
    MDM2_chronos = chronos[,"MDM2"]
    )

ggplot(plot_data, aes(TP53_status, MDM2_chronos, fill = TP53_status)) +
  geom_boxplot() +
  theme(legend.position = "bottom")
```



2.5 (Bonus exercise)

Is the observed difference significant? Try to test this using a `t.test()` between the two sets of values. Is the p-value smaller than 0.05?

```
t_test <- t.test(plot_data[TP53_status == "TP53 mutant", MDM2_chronos],
       plot_data[TP53_status == "TP53 WT", MDM2_chronos])


t_test$p.value < 0.05
```

```
[1] TRUE
```